

The National Information Ecosystem

Evidence, institutions and accountable AI readiness

Candidate lesson presentation | v0.1



Learning outcomes

Map

Define the ecosystem boundary and its actors

Trace

Follow information from source to decision

Assess

Evaluate quality, authority and safeguards

Decide

Screen an AI opportunity using evidence

The information ecosystem

People and institutions

Data and information assets

Rules, processes and technology

Authority and accountability

Flows and transformations

Decisions and public value

An ecosystem explains how these elements interact within a stated boundary.

Three nested boundaries

National information ecosystem

Official statistics and the national statistical system

Product or use case

State the boundary before assessing readiness

Roles and institutional authority

Producer

Creates or compiles
information

Methodological authority

Defines standards
and classifications

Custodian

Maintains records
and access

Coordinator

Aligns actors and
resolves gaps

One institution may hold several roles. Each role still needs explicit authority and accountability.

Information assets

1

Administrative records

Registers, transactions and service records

2

Surveys and censuses

Designed observations with stated populations

3

Geospatial and sensor data

Location, imagery and environmental measurements

4

Reference information

Classifications, metadata and business rules

Each asset requires an owner, purpose, coverage, timing, quality record and access rule.

Information flows and transformations

Document every handoff



Actor, timing, format, control and failure

A flow becomes auditable when responsibility and evidence follow the data.

Quality and safeguards

Statistical quality

Relevance
Accuracy
Timeliness
Coherence
Accessibility

Institutional safeguards

Authority
Confidentiality
Security
Auditability
Human responsibility

Both sides must support the intended decision

Evidence labels



Evidenced

Supported by a document, dataset, system record or validated observation



Reported

Stated by a knowledgeable respondent but not independently confirmed



Unknown

Information unavailable, conflicting or still under review

Do not convert reported information into evidence by repetition.

Fictional case: Republic of Kintaba

A country with multiple producers, fragmented flows and pressure for faster decisions

12 actors

10 assets

8 flows

8 evidence records

6 risks

5 AI opportunities

The case is synthetic. The institutional problems are realistic.

Exercise sequence

1

**Map actors and
authority**

2

Inventory assets

3

**Trace a critical
flow**

4

**Screen an AI
opportunity**

Record evidence, missing information and responsible follow-up at every stage.

AI opportunity screening

Purpose Which decision or service would improve?

Baseline What currently happens?

Data Which assets support the task?

Quality Which limitations affect validity?

Authority Who may authorize use?

Decision Proceed, improve first or stop?

Common failure patterns

Inventory without purpose	Lists assets but cannot connect them to decisions
Mandate assumed	Treats access or ownership as authority
Quality averaged	Hides a critical weakness inside a composite score
AI selected first	Forces ecosystem evidence to justify a predetermined tool

A credible diagnostic makes disagreement and uncertainty visible.

From diagnostic to action plan

0–90 days

**Confirm
mandates and
evidence gaps**

3–6 months

**Repair priority
flows and quality
controls**

6–12 months

**Run controlled
pilots with
assurance**

Ongoing

**Review
outcomes, risks
and public value**

Assign an owner, evidence requirement, deadline and decision point to each action.

Assessment evidence

A defensible submission demonstrates

- ✓ A stated and justified boundary
- ✓ Actors and authority supported by evidence
- ✓ A traceable critical information flow
- ✓ A realistic AI screening decision
- ✓ Clear treatment of uncertainty and safeguards

The assessment rewards reasoning and evidence, not optimism about AI.

One method across multiple public decisions

Agriculture, education, health and habitat depend on different evidence, but each requires explicit authority, quality and safeguards.

The next section examines the Kintaba evidence package.



Kintaba evidence package

12

actors

10

assets

8

flows

8

evidence records

6

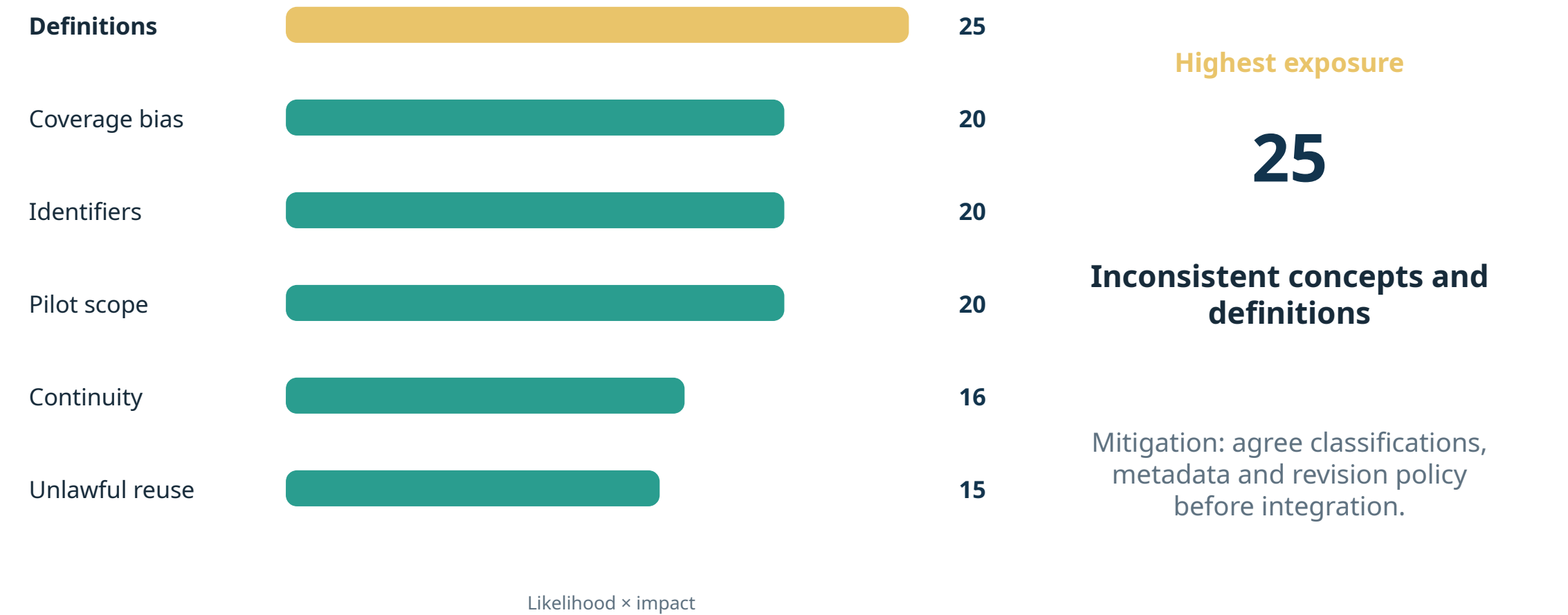
material risks

5

AI opportunities

The files connect institutional claims to reproducible analytical outputs.

Material risk exposure



Five candidate AI opportunities



Disposition by opportunity

Crop-condition nowcast

Controlled pilot only

Estimate reconciliation

Metadata and provenance first

Holding-frame prioritization

Research prototype

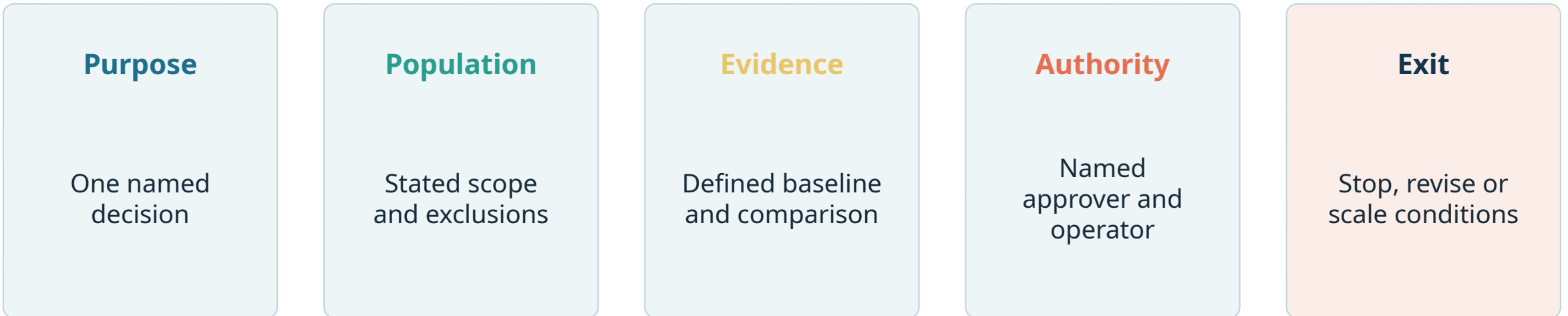
Crop-stress detection

Controlled pilot only

Irrigation gap detection

Improve source data before automation

Controlled pilot boundary



A pilot tests a bounded claim. It does not authorize general deployment.

Statistical assurance record

Baseline

Current process and error profile

Validity

Metrics tied to the intended use

Subgroups

Performance across relevant domains

Drift

Monitoring frequency and triggers

Provenance

Source and transformation history

Reproducibility

Code, versions and parameters

Review

Independent challenge and approval

Release

Scope, limitations and revision note

The assurance record preserves the evidence needed to reproduce and challenge the result.

Institutional action priorities

Now

Confirm mandates, classifications and evidence gaps

Next

Repair priority flows, identifiers and metadata

Then

Run bounded pilots with statistical assurance

Always

Review risk, public value and accountability

The action plan strengthens the ecosystem whether or not an AI pilot proceeds.

A defensible AI decision begins with trustworthy official statistics

PROCEED

Evidence supports a bounded pilot

IMPROVE FIRST

Material gaps can be repaired

STOP

Purpose, authority or safeguards are insufficient

Next: complete the Kintaba laboratory and preserve the evidence record.